

AVIRUP DAS

M.Sc in Big Data Analytics

RKMVERI, Belur Math, West Bengal, India

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🌐 Portfolio



PROJECTS

- **Multi-Agent Smart Shopping Intelligence – Agentic RAG-Based Product Recommendation System**
LangGraph | LangChain | Groq LLM | SerpAPI | Agentic RAG | Generative AI [GitHub] Jan 2026
 - Built a **multi-agent GenAI shopping assistant** using LangGraph for query understanding, retrieval, ranking, and recommendations.
 - Integrated **Google Shopping (SerpAPI)** for real-time product search with automated data cleaning and fallback handling.
 - Developed a **budget-aware recommender** supporting natural language price ranges with deduplication and diversity-aware ranking.
- **NewsSense AI – Agentic RAG-Powered News Summarizer & Fact-Checker (On-going)**
LangChain | RAG | AI Agents | Generative AI [GitHub] Jan 2026
 - Designing a **Generative AI system** leveraging LangChain, RAG, and AI agents to analyze, summarize, and fact-check news articles.
 - Implementing an **agentic workflow** where a planner agent autonomously decides when to summarize, fact-check (Wikipedia/NewsAPI), and run sentiment analysis.
- **Multimodal-Hierarchical-CNN-for-Sun-Salutation-Pose-Classification**
OpenCV | CNN | CNN+LSTM | 3D CNN [GitHub] June 2025 - Nov 2025
 - Developed a real-time system for Sun Salutation guidance, leveraging **MediaPipe** and a **3d Quadtree CNN** for accurate pose recognition.
 - Developed **3D quadtree CNN** and VIT for capturing the spatio-temporal features.
- **Speech Emotion Recognition**
Deep Learning | MFCC | ANN | CNN [GitHub] Jan 2025 - May 2025
 - Developed a deep learning-based Speech Emotion Recognition system using ANNs and CNNs on RAVDESS, CREMA-D, and SAVEE datasets.
 - Implemented extensive MFCC feature extraction and data augmentation to enhance model generalization and robustness.
- **Graph-Based Recommendation System**
Scikit-learn | TF-IDF | Cosine Similarity | Neo4j [GitHub] Jan 2025 - Apr 2025
 - Built a graph-based recommendation engine combining **content-based**, **collaborative**, and **session-based** filtering.
 - Modeled user-item interactions in **Neo4j** using Cypher and cosine similarity for personalized recommendations.
 - Applied **TF-IDF** on genre metadata and user/item/session similarity to improve recommendation accuracy.

COURSEWORK

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|--------------------------------------|-------------------------------|
| • Machine Learning | • Finance and Econometrics |
| • Deep Learning and NLP | • Time Series |
| • Computer Vision | • Deep Reinforcement Learning |
| • Data Structures and Algorithms | • DBMS |
| • Statistics | • Mining of Massive Dataset |
| • Probability and Stochastic Process | |

ACHIEVEMENTS

- Gold Medal (1st Class 4th Position) Graduation, RKMRC, 2024
- Top 5% in NPTEL, Joy of Computing with Python
- Ray and Martin Scholarship-Awarded for WBBSE 2019
- NMMSE Scholarship-Awarded for qualifying the NMMSE 2016

EXPERIENCE

Summer Research Intern

FLAME UNIVERSITY

📅 June 2025 - July 2025 📍 Pune, India

Real-time interactive system for the "**Sun Salutation Pose Guide**". under the supervision of Dr. Chiranjoy Chattopadhyay.

EDUCATION

- **Ramakrishna Mission Vivekananda Educational and Research Institute, Howrah**
M.Sc in Big Data Analytics
📅 2024 – Present (Till Sem-1) CGPA: 8.10
- **Ramakrishna Mission Residential College.**
B.Sc in Mathematics
📅 2021 – 2024 CGPA: 9.10
- **A.C Institution**
Higher Secondary (10+2)
📅 2019 – 2021 Score: 88.2%
- **Birampur High School**
Secondary (10)
📅 2013 – 2019 Score: 95.8%

TECHNICAL SKILLS

- **Programming Languages:** Python, C, R
- **Libraries & Frameworks:** Pytorch, OpenCV, scikit-learn, Seaborn, SQL, Neo4j, NumPy, Pandas, Matplotlib
- **Operating System :** Windows, Linux (Ubuntu)

ACTIVITY

- **Placement Coordinator, RKMVERI**
– Manage Placement Cell for the Batch of 2024-26
- **Chegg (Remote) - Subject Matter Expert (Mathematics)**
– Provided detailed step-by-step solutions for Computer Science problems
- **Fest Organiser**
– Core Committee Member **Perceptron 2025** Auction Group
– Core Organizing Team Member **Infinity 2024** at RKMRC

HOBBY

- Reading Books , Watching Movies , Playing Cricket